

CPEN 416 / EECE 5710

Lecture 06: Multi-qubit measurements, the Bell basis, and superdense coding

Tuesday 23 September 2025

Announcements

Office hour updates:

- no hour tomorrow
- Friday hour 13:00 - 13:50
- will be avail. Tues. 30 Sept. during lecture time

- Midterm in class on Thursday 2 Oct (see PrairieLearn for details)
- Quiz 3 today
- Tutorial Assignment 1 due Friday 23:59
- Tutorial 3 supplement w/ practice problems now available

Last time

We used the tensor product to construct multi-qubit states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

$$\begin{aligned}
 |\psi\rangle \otimes |\varphi\rangle &= (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) \\
 &= \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}
 \end{aligned}$$

Last time

$$i = e^{i\pi/2}$$

$$= e^{i\pi/4} e^{i\pi/4}$$

$$|0\rangle \xrightarrow{B} |b_1\rangle$$

$$|1\rangle \xrightarrow{B} |b_2\rangle$$

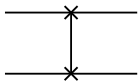
We saw examples of two-qubit gates:

$$|\psi\rangle = \begin{pmatrix} \frac{2i}{\sqrt{5}} \\ \frac{e^{i\pi/3}}{\sqrt{5}} \end{pmatrix}$$

$$|b_2\rangle = \begin{pmatrix} \frac{1}{2}e^{-i\pi/4} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$$

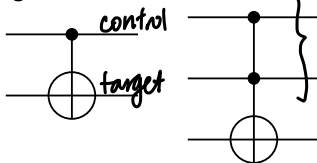
$$\langle b_2 | \psi \rangle = \left(\frac{1}{2}e^{i\pi/4} \quad \frac{\sqrt{3}}{2} \right) \begin{pmatrix} \frac{2i}{\sqrt{5}} \\ \frac{e^{i\pi/3}}{\sqrt{5}} \end{pmatrix} \text{ if control is state } |1\rangle = \text{apply } X \text{ to target}$$

$$\Rightarrow |\langle b_2 | \psi \rangle|^2 = \text{Pr}(|b_2\rangle)$$



SWAP

CNOT



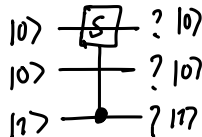
Toffoli

$$|b_1\rangle \xrightarrow{B^\dagger} |0\rangle$$

$$|b_2\rangle \xrightarrow{B^\dagger} |1\rangle$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$S|0\rangle = |0\rangle$$

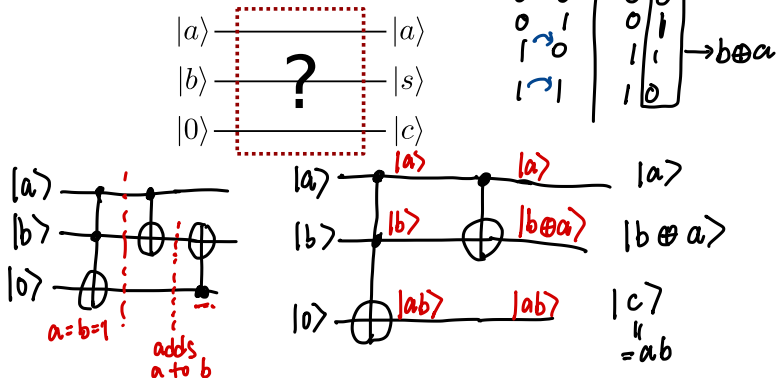


$$|10\rangle \xrightarrow{CS} i|10\rangle$$

Last time

X, CNOT, TOF can be used to create Boolean arithmetic circuits.

Exercise: use a combination of these gates to create a quantum version of a one-bit adder. Test it in PennyLane on all possible bit values of a and b .



Learning outcomes

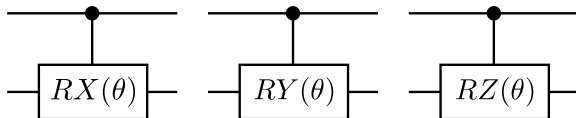
- apply gates controlled on the state of multiple qubits
- define and give examples of entangled states
- measure multi-qubit systems in the computational basis and Bell basis
- leverage entanglement to implement superdense coding

Example: controlled rotations (RX , RY , RZ)

We can control arbitrary Pauli rotations:

$$CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ 0 & 0 & \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

Circuit elements:



PennyLane: `qml.CRX`, `qml.CRY`, `qml.CRZ`

Image credit: Codebook node I.14

Controlled- U

There is a pattern here:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad CRY(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ 0 & 0 & \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

Z
 $RY(\theta)$

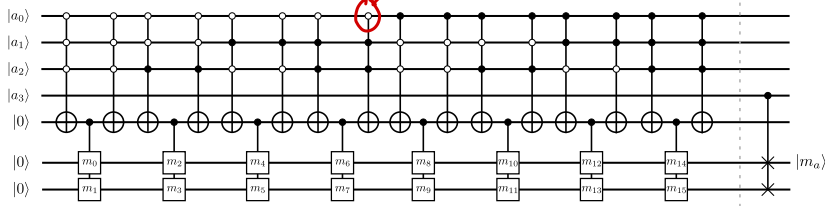
More generally,

$$CU = \begin{pmatrix} I_2 & 0 \\ 0 & U \end{pmatrix}, \quad U = \begin{pmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{pmatrix}$$

Controlled unitary operations

Any unitary operation can be turned into a controlled operation, controlled on any state.

*if qubit in $|0\rangle$
do something...*



Most common controls are controlled-on- $|1\rangle$ (filled circle), and controlled-on- $|0\rangle$ (empty circle).

Controlled unitary operations with `qml.ctrl`

Remember `qml.adjoint`:

```
@qml.qnode(dev)
def my_circuit():
    qml.S(wires=0)
    qml.adjoint(qml.S)(wires=0)
    return qml.sample()
```

A similar *transform* allows us to perform arbitrary controlled operations (or entire quantum functions)!

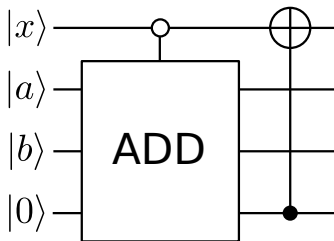
```
@qml.qnode(dev)
def my_circuit():
    qml.S(wires=0)
    qml.ctrl(qml.S, control=1)(wires=0)
    return qml.sample()
```

targets
↑

Controlled unitary operations with `qml.ctrl`

Take home exercise: using your adder from earlier, implement a circuit in PennyLane that uses `qml.ctrl` to

- perform addition if the control qubit is in state $|0\rangle$
- flip the state of the control qubit if the carry bit is 1

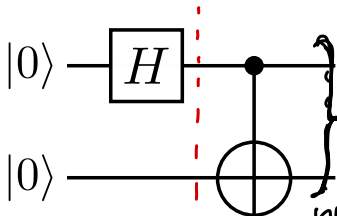


Controlling on superpositions

Exercise: What does CNOT do when the control qubit is in a superposition? Determine the combined output state of this circuit, then try to express it as a tensor product of two single-qubit states.

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$\neq U_1 \otimes U_2$



$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$| \psi \rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

normalized ✓
⇒ entangled!

$$\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= (\alpha |0\rangle + \beta |1\rangle) \otimes (\gamma |0\rangle + \delta |1\rangle)$$

No!

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle$$

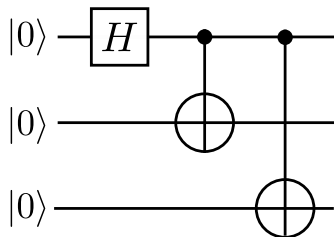
$$= \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

separable ↑

Entanglement

CNOT is an **entangling gate**!

Entanglement generalizes to more than two qubits:



Exercise: Express the output state of this circuit in the computational basis.

Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

If we measure in the computational basis, the outcome probabilities are:

- $|\alpha|^2 = |\langle 00|\psi\rangle|^2$ for $|00\rangle$
- $|\beta|^2 = |\langle 01|\psi\rangle|^2$ for $|01\rangle$
- ...

Exercise: what is the probability of the qubits being in state $|110\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?

Multi-qubit measurement outcome probabilities

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

We can also measure *just one qubit*:

- The probability of the first qubit being in state $|0\rangle$ is $|\alpha|^2 + |\beta|^2$
- The probability of the second qubit being in state $|1\rangle$ is $|\beta|^2 + |\delta|^2$

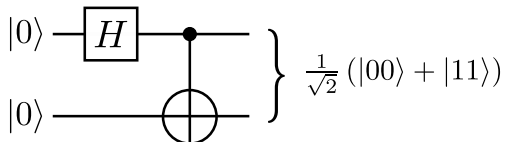
Exercise: what is the probability of the second qubit being in state $|0\rangle$ after measuring $(H \otimes SX \otimes I)|000\rangle$ in the computational basis?

Bell states

Remember how we created

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle),$$

from the $|00\rangle$ state:



Bell states

Exercise: Apply the same circuit to the other 3 computational basis states? What is special about these four states?

$$\begin{array}{c} |0\rangle \\ |0\rangle \end{array} \begin{array}{c} \boxed{H} \\ \oplus \end{array} \rightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\begin{array}{c} |0\rangle \\ |1\rangle \end{array} \begin{array}{c} \boxed{H} \\ \oplus \end{array}$$

$$|\Psi_{01}\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$\begin{array}{c} |1\rangle \\ |0\rangle \end{array} \begin{array}{c} \boxed{H} \\ \oplus \end{array}$$

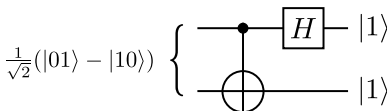
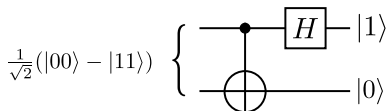
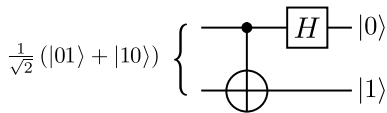
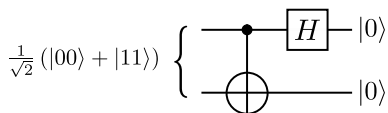
$$|\Psi_{10}\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\begin{array}{c} |1\rangle \\ |1\rangle \end{array} \begin{array}{c} \boxed{H} \\ \oplus \end{array}$$

$$\begin{aligned} |\Psi_{11}\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \end{aligned}$$

The Bell basis

We can measure in this basis by applying the adjoint of the circuit:



Two famous quantum algorithms, **superdense coding** and **teleportation** work by performing a measurement in the Bell basis.

Next time

Content: *superdense coding*

- Quantum teleportation

Action items:

- 1 Tutorial assignment 1 due Friday 23:59
- 2 Study for midterm

Recommended reading:

- Codebook nodes IQC, SQ, MQ; Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3